# THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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## ANSWER KEY & MARKING SCHEME SET A

Chapter 9: **Some Applications of Trigonometry** | Class 10 Mathematics | Maximum Marks: 40

### SECTION — A (Objective Type) — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** $5\sqrt{3}$ m | $10\sin 60^\circ$ |
| 2 | **(a)** 12 m | $\tan 45^\circ = 1$ |
| 3 | **(a)** $30^\circ$ | $\tan\theta = \frac{1}{\sqrt{3}}$ |
| 4 | **(b)** angle of depression | Definition |
| 5 | **(b)** Incorrect — it increases | $\tan\theta = \frac{h}{d}$ increases as $d$ decreases |
| 6 | **(a)** $40\sqrt{3}$ m | $\frac{60}{\sin 60^\circ} = \frac{120}{\sqrt{3}}$ |
| 7 | **(b)** Student B is correct | $28.5 + 1.5 = 30$ m |
| 8 | **(a)** $30^\circ$ | $\tan\theta = \frac{1}{\sqrt{3}}$ |
| 9 | **(a)** $20\sqrt{3}$ m | $\frac{60}{\tan 60^\circ}$ |
| 10 | **(b)** $45^\circ$ | $\tan\theta = 1$ |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | $\sin 30^\circ = \frac{h}{20} \Rightarrow h = 10$ m | 1+1 |
| 12 | $\tan 30^\circ = \frac{10}{d} \Rightarrow d = 10\sqrt{3} \approx 17.32$ m | 1+1 |
| 13 | $\sin 30^\circ = \frac{1.5}{l} \Rightarrow l = 3$ m | 1+1 |
| 14 | $\tan\theta = \frac{20}{20} = 1 \Rightarrow \theta = 45^\circ$. Claim is **false**. | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | $h\sqrt{3} - \frac{h}{\sqrt{3}} = 40 \Rightarrow \frac{2h}{\sqrt{3}} = 40 \Rightarrow h = 20\sqrt{3} \approx 34.64$ m | 1+1+1 |
| 16 | Height above eye $= 28.5$ m. Distance $= 28.5\left(\sqrt{3} - \frac{1}{\sqrt{3}}\right) = \frac{57}{\sqrt{3}} = 19\sqrt{3} \approx 32.9$ m | 1+1+1 |
| 17 | $h = x\sqrt{3} = \frac{80-x}{\sqrt{3}} \Rightarrow x = 20$; $h = 20\sqrt{3} \approx 34.64$ m; distances 20 m and 60 m | 1+1+1 |
| | **OR:** $h = x\sqrt{3} = \frac{x+20}{\sqrt{3}} \Rightarrow x = 10$; width 10 m, height $10\sqrt{3} \approx 17.32$ m | 1+1+1 |
| 18 | Pedestal $h$, distance $h$ $(45^\circ)$. $\sqrt{3} = \frac{h+1.6}{h} \Rightarrow h = \frac{1.6}{\sqrt{3}-1} = 0.8(\sqrt{3}+1) \approx 2.19$ m | 1+1+1 |
| | **OR:** Distances $75\sqrt{3}$ and 75 m ⇒ $75(\sqrt{3}-1) \approx 54.9$ m | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | Horizontal distances $\frac{3600\sqrt{3}}{\sqrt{3}} = 3600$ and $3600\sqrt{3} \cdot \sqrt{3} = 10800$ m; covered 7200 m in 30 s ⇒ 240 m/s $= 864$ km/h | 2+2+1 |
| | **OR:** Distances $h\sqrt{3}$ and $\frac{h}{\sqrt{3}}$; in 6 s it covers $\frac{2h}{\sqrt{3}}$. Remaining $\frac{h}{\sqrt{3}}$ takes **3 s** | 2+2+1 |
| 20 | (ii) $80\sqrt{3} \approx 138.56$ m and $\frac{80}{\sqrt{3}} \approx 46.19$ m (iii) $\frac{320}{\sqrt{3}} = \frac{320\sqrt{3}}{3} \approx 184.75$ m (iv) The one seeing it at $60^\circ$ — larger angle, smaller distance | 1+2+1+1 |
| | **OR:** (ii) $\tan 45^\circ = \frac{10}{d} \Rightarrow d = 10$ m (iii) part above window $= 10\tan 60^\circ = 10\sqrt{3}$; height $= 10(1+\sqrt{3}) \approx 27.32$ m | 1+2+2 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

## ANSWER KEY & MARKING SCHEME — SET B

Chapter 9: **Some Applications of Trigonometry** | Class 10 Mathematics | Maximum Marks: 40

### SECTION — A (Objective Type) — 10 × 1 = 10

| Q. | Correct Option | Brief Reasoning |
|---|---|---|
| 1 | **(a)** 5 m | $\frac{2.5}{\cos 60^\circ}$ |
| 2 | **(c)** $60^\circ$ | $\tan\theta = \frac{6}{2\sqrt{3}} = \sqrt{3}$ |
| 3 | **(a)** 50 m | $\tan 45^\circ = 1$ |
| 4 | **(a)** $100\sqrt{3}$ m | $100\tan 60^\circ$ |
| 5 | **(b)** Incorrect — they are equal (alternate angles) | Horizontal lines are parallel |
| 6 | **(a)** 50 m | $100\sin 30^\circ$ |
| 7 | **(a)** Student A is correct | $\tan\theta = \frac{1}{\sqrt{3}}$ |
| 8 | **(b)** decreases | $\tan\theta = \frac{h}{\text{shadow}}$ |
| 9 | **(a)** 80 m | $\tan 45^\circ = 1$ |
| 10 | **(b)** $\frac{20}{\sqrt{3}}$ m | $20\tan 30^\circ$ |

### SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 11 | $\tan\theta = \frac{12}{4\sqrt{3}} = \sqrt{3} \Rightarrow \theta = 60^\circ$ | 1+1 |
| 12 | $\tan 60^\circ = \frac{10}{d} \Rightarrow d = \frac{10}{\sqrt{3}} = \frac{10\sqrt{3}}{3} \approx 5.77$ m | 1+1 |
| 13 | Height $= 3.7$ m; $l = \frac{3.7}{\sin 60^\circ} = \frac{7.4}{\sqrt{3}} \approx 4.27$ m | 1+1 |
| 14 | Height above his eyes $= 20\tan 45^\circ = 20$ m; tower $= 21.8$ m. Claim is **false** (eye height ignored). | 1+1 |

### SECTION — C (Short Answer) — Answers — 4 × 3 = 12

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 15 | (i) $40\tan 30^\circ = \frac{40}{\sqrt{3}} \approx 23.09$ m (ii) top of tank at 40 m ⇒ depth $= 40 - 23.09 \approx 16.91$ m | 1½+1½ |
| 16 | Distances $\frac{100}{\sqrt{3}} \approx 57.74$ m and 100 m; moved $42.26$ m in 2 min ⇒ $\approx 21.13$ m/min | 1+1+1 |
| 17 | Let height $H$, distance $d = H$. $H - 8 = \frac{H}{\sqrt{3}} \Rightarrow H = \frac{8\sqrt{3}}{\sqrt{3}-1} = 4(3+\sqrt{3}) \approx 18.93$ m; distance $\approx 18.93$ m | 1+1+1 |
| | **OR:** $d = \frac{50}{\tan 30^\circ} = 50\sqrt{3}$; hill $= d\tan 60^\circ = 150$ m | 1+1+1 |
| 18 | $d = 10\sqrt{3} \approx 17.32$ m; flagstaff $= d - 10 = 10(\sqrt{3}-1) \approx 7.32$ m | 1+1+1 |
| | **OR:** $h = x\sqrt{3} = \frac{x+40}{\sqrt{3}} \Rightarrow x = 20$ m (width); $h = 20\sqrt{3} \approx 34.64$ m | 1+1+1 |

### SECTION — D (Long Answer) — Answers — 2 × 5 = 10

| Q. | Expected Answer / Value Points | Marks |
|---|---|---|
| 19 | (i) $\frac{60}{\sqrt{3}} = 20\sqrt{3} \approx 34.64$ m (ii) $60 - 20\sqrt{3} \approx 25.36$ m (iii) False — top is seen below the horizontal, so the tower is shorter | 2+2+1 |
| | **OR:** $d = \frac{10}{\tan 30^\circ} = 10\sqrt{3} \approx 17.32$ m; height above eye $= d\tan 60^\circ = 30$ m ⇒ hill $= 40$ m | 2+2+1 |
| 20 | (i) $\frac{50}{\sqrt{3}} \approx 28.87$ m (ii) $50\sqrt{3} \approx 86.6$ m (iii) $\frac{100}{\sqrt{3}} \approx 57.74$ m (iv) A (larger angle) | 1+1+2+1 |
| | **OR:** (i) $\frac{h}{\sqrt{3}}$ and $h$ (ii) $h\left(1+\frac{1}{\sqrt{3}}\right) = \frac{100(\sqrt{3}+1)}{\sqrt{3}} \Rightarrow h = 100$ m | 2+3 |

**Evaluator's note:** Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.